

SQL Case Study: Hospital Wait Time Investigation

Business Analysis project — Vaishnavi Suryawanshi

Business Question

Wait times across hospital departments vary significantly. This analysis investigates whether department, staffing levels, or registration method best explains that variation, in order to recommend where an operational fix would have the most impact.

Dataset

- 30 patient-visit records
 - 3 departments: Cardiology, Orthopedics, Radiology
 - Columns: patient_id, department, wait_time_min, staff_available, day, registration_type
 - Reference table: department_lead (department — head_doctor)
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Analysis 1 — Wait Time by Department

Question it answers: Which departments have the worst wait times?

```
SELECT department, AVG(wait_time_min) AS avg_wait_time
FROM patient_data
GROUP BY department;
```

Result: Cardiology ≈ 28.3 min · Radiology ≈ 44.6 min · Orthopedics ≈ 56.2 min

Insight: Orthopedics has the longest average wait — nearly double Cardiology's — making it the top priority for operational review.

Analysis 2 — Wait Time by Registration Type

Question it answers: Does digital registration actually reduce wait time?

```
SELECT registration_type, AVG(wait_time_min) AS avg_wait_time
FROM patient_data
GROUP BY registration_type;
```

Result: Manual ≈ 50.1 min · Digital ≈ 26.4 min

Insight: Digital registration cuts average wait time by roughly 47%, but is currently only used in Cardiology — expanding it to other departments is a clear, low-effort opportunity.

Analysis 3 — Departments Needing Attention

Question it answers: Which departments cross an acceptable wait-time threshold?

```
SELECT department, AVG(wait_time_min) AS avg_wait_time
FROM patient_data
GROUP BY department
HAVING AVG(wait_time_min) > 40;
```

Result: Radiology (≈ 44.6 min) and Orthopedics (≈ 56.2 min) both exceed a 40-minute average; Cardiology does not.

Insight: Two of three departments cross the threshold — this isn't an isolated problem in one department, it's a broader operational pattern.

Analysis 4 — Enriching Data with Department Leadership

Question it answers: Who owns accountability for each department's performance?

```
SELECT d.head_doctor, AVG(p.wait_time_min) AS avg_wait_time
FROM patient_data p
JOIN department_lead d ON p.department = d.department
GROUP BY d.head_doctor;
```

Result: Dr. Mehta (Cardiology) ≈ 28.3 min · Dr. Kulkarni (Radiology) ≈ 44.6 min · Dr. Rane (Orthopedics) ≈ 56.2 min

Insight: Joining performance data to department ownership turns a descriptive finding into an actionable one — each result now has a named stakeholder.

Analysis 5 — Data Quality Check

Question it answers: Can this analysis be trusted — is any department missing a match?

```
SELECT p.department, d.head_doctor
FROM patient_data p
LEFT JOIN department_lead d ON p.department = d.department
WHERE d.head_doctor IS NULL;
```

Result: Empty result set.

Insight: Every department in the patient data has a matching entry in the lead table — confirming the joins used in Analysis 4 are reliable, with no silently dropped or unmatched rows.

Analysis 6 — Within-Department Ranking

Question it answers: Which specific patients are outliers within their own department, not just on average?

```
SELECT patient_id, department, wait_time_min,  
       RANK() OVER (PARTITION BY department ORDER BY wait_time_min  
DESC) AS dept_rank  
FROM patient_data;
```

Result (example): P009 ranks #1 longest wait within Orthopedics at 65 minutes.

Insight: Averages alone hide outliers. Ranking within each department surfaces the specific patient cases most worth investigating individually, rather than treating each department as a single number.

Overall Recommendation

Process design — not staffing levels or patient volume — is the primary driver of wait time variation. Prioritizing digital registration rollout beyond Cardiology, starting with Orthopedics, is projected to reduce average wait time by 20–30% overall.